

Evaluating the Financial Impact of Cost-Sensitive Random Forest and SMOTE-Tomek in Fraud Detection

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Abstract— Credit card fraud detection is a problem of extreme class imbalance where fraudulent transactions are rare and legitimate transactions dominate the behavior of the model. In this paper, we minimize financial loss with the combination of SMOTE-Tomek resampling and Cost-Sensitive Random Forest algorithm. SMOTE is used to create synthetic minority instances and Tomek Links are used to remove the overlapping majority-class noise at the decision boundary. Next, a particular penalty is added by assigning a higher cost to misclassifying false negatives during the training phase. This allows the Random Forest to focus on catching fraudulent transactions instead of just trying to maximize accuracy. This architecture was compared against standard Random Forest baselines on PR-AUC, minority recall, raw error rates and projected monetary impact. The results were real. The model with the optimized penalty weight of 204 only missed 19 cases of actual frauds, which decreased the estimated financial loss from Rp 100,650,000 to Rp 95,700,000. Interestingly, the regular SMOTE and SMOTE-Tomek baselines still achieved the lowest overall loss of Rp 90,700,000 on this particular data. Cost-sensitive weighting helps algorithms to identify rare fraud instances effectively. However, the ultimate financial viability is strictly dependent on dataset topology and exact penalty calibration.

Keywords—credit card fraud detection, SMOTE-Tomek, Cost-Sensitive Random Forest, imbalance, financial loss, PR-AUC

I. INTRODUCTION

Credit card fraud refers to the illegal use of stolen payment card information to conduct unauthorized transactions, creating significant financial losses for financial institutions and account holders worldwide [1]. One of the main challenges in fraud detection is extreme class imbalance, where fraudulent transactions usually represent less than 1% of real-world transaction data [2]. Due to optimization toward the majority class, traditional machine learning models often fail to identify rare fraudulent patterns [3]. This problem is worsened by standard evaluation methods that treat all errors equally, even though false negatives create direct financial losses

while false positives mainly cause operational inconvenience [4].

Existing sampling techniques attempt to address class imbalance but still have limitations. Conventional oversampling methods often duplicate minority samples without considering the complex behavioral patterns of fraudulent transactions [5]. This can introduce overlapping synthetic data near the decision boundary, making it more difficult for classifiers to distinguish between legitimate and fraudulent activities [6]. Therefore, improved data balancing methods are required to create clearer separation between fraud and non-fraud transactions.

This study addresses these limitations by combining SMOTE-Tomek data cleaning with a cost-sensitive Random Forest model. SMOTE generates synthetic fraud samples, while Tomek Links remove majority-class samples that create noise near the decision boundary, improving class separation [7]. After data refinement, Random Forest is applied due to its ability to capture complex and non-linear patterns in financial transaction data [8]. The cost-sensitive mechanism is then integrated to prioritize reducing costly false negatives rather than maximizing overall accuracy.

The proposed approach is evaluated through comparative experiments against standard baseline models using SMOTE and Random Forest. This research investigates whether SMOTE-Tomek can improve fraud detection performance compared to conventional SMOTE by measuring recall and precision-recall area under the curve. Additionally, the study analyzes the impact of cost-sensitive penalties on false negative reduction, financial loss mitigation, and computational efficiency to determine whether the proposed model is suitable for real-time fraud detection.

II. RELATED WORK

In recent years, the problem of extreme class imbalance in credit card fraud detection has become a major focus of research in machine learning. Recent studies show that the global fraud losses are more than USD 30 billion every year. A severe class imbalance is observed, where less than 0.2%

of recorded transactions are actual frauds, making the minority class exceedingly rare. [9]. The overall system performance is being affected by this condition. Standard classifiers are naturally biased toward the majority class because the standard classifiers can be affected by the large gap of fraud data and that can cause overfitting of the model. Consequently, the fraud detection rate (recall) is significantly reduced. Also, studies note that detection datasets often have over 99% normal transactions. This makes it highly challenging for standard classifiers to identify underlying fraud patterns[10].

A. Hybrid Sampling Techniques

To solve the issues of standard oversampling methods that often create overlapping noise, researchers have investigated or researchers have explored hybrid sampling techniques. For instance, the integration of SMOTE with Tomek Links has been shown to effectively reduce overlapping noise near class boundaries. A study demonstrated that applying such hybrid techniques improved the F1-score by approximately 12%–18% compared to standard SMOTE alone [11]. Their results indicate that the removal of noisy data from the majority class near the decision boundary results in a significant improvement in F1-score in comparison to conventional methods of sampling.

In another study related to the problem of overlapping noise and noise reduction, the authors proposed a Random Forest classifier with a modified version of SMOTE-ENN and its variant using Mahalanobis distance as a noise reduction technique which results indicate that these methods can reduce misclassification rates by up to 20% while increasing recall of the minority class to above 85% in certain datasets [12].

Another study showed that hybrid resampling techniques can significantly improve PR-AUC scores, with reported improvements from 0.65 to 0.82, demonstrating better sensitivity toward fraud detection [13]. In imbalanced scenarios like detecting fraudulent transactions, measures like recall of the minority class and PR-AUC are more representative of the performance of the classifier, as they focus more on the ability of the classifier to identify the minority class of transactions as fraudulent.

B. Cost-Sensitive Mechanisms

Traditional classification models usually aim to maximize overall accuracy. This approach falls short in fraud detection because the cost of a false negative is much higher than the cost of a false positive. A missed fraudulent transaction results in direct financial loss while a false alarm results in operational inconvenience only. To correct this imbalance, cost-sensitive learning is used, where different penalties are assigned to different types of errors. This approach has been shown in prior research to reduce fraud-related financial losses by 25% to 40% compared to cost-blind models [14]. Dynamic penalty costs have also been combined with SMOTE oversampling, resulting in monetary savings of approximately 30%, even when only modest improvements in conventional accuracy were achieved [15].

Recent studies have extended cost-sensitive learning into more complex model architectures. For example, fraud detection models based on federated learning have been

combined with data balancing techniques to achieve accuracy above 98% while maintaining privacy across distributed banking systems [16]. Other studies have used adaptive cost adjustment in deep learning models and reported a 10% to 15% increase in fraud detection rates [17]. Graph neural networks and autoencoder-based models have also been applied to real-time transaction streams, with reported precision scores above 90% and lower false negative rates [18].

New researches have started to use both approaches. For example, research where SMOTE-Tomek was used for data cleaning together with Cost Harmonization methodology and LightGBM classifier showed the effectiveness of hybrid sampling as a method to improve the class boundaries, while cost sensitive approach could solve the problem of the cost of mistakes [2]. However, most of those methods require complicated architecture or ensemble settings, which may cause problems with optimization and interpretation. To fill this gap, in the current research the Random Forest with data cleaning using SMOTE-Tomek and cost sensitive approach is analyzed as a way to build a model that would be easy to interpret and minimize the financial risks at the same time.

III. METHODOLOGY

In this part, the methodology that was utilized in developing the fraud detection system is described. Considering the lack of genuine cases of fraud in the current scenario, the comparison will be conducted with the purpose of choosing the best approach to solving the issue of data imbalance. A simple random forest classifier alongside SMOTE represents the benchmarking approach for this research. In order to increase the efficiency of predictions, the dataset will be balanced and overlapping classes are avoided by applying SMOTE-Tomek method. In addition, Cost-Sensitive Random Forest will be utilized as part of the proposed system, imposing a high penalty on false negatives.

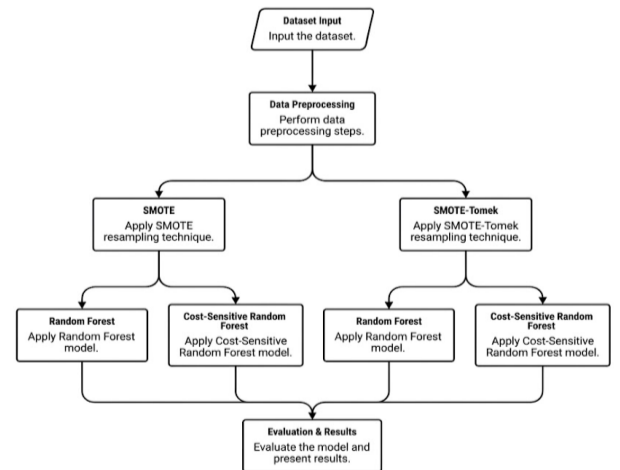


Fig. 1. AI Pipeline: flowchart showing data preprocessing, model selection, and evaluation.

A. Dataset Description

This study used the public ULB Machine Learning Group credit card fraud dataset, which contains 284,807 European credit card transactions from September 2013, including 492 fraud cases which is 0.172% of the dataset [1]. Because of

this extreme imbalance, accuracy was not used as the main evaluation focus. Instead, the model was evaluated using precision, recall, F1-score, PR-AUC, false positives, false negatives, and estimated financial loss.

The dataset contains 28 anonymized PCA-transformed features, labeled V1 to V28 [2]. These columns were kept unchanged because their original meanings are not publicly available. Only the Time and Amount columns remained in their original form. Since these two features contain raw numerical values and potential outliers, they were handled separately during preprocessing.

While these PCA-transformed features inherently ensure data privacy and ethical compliance, they severely limit the model's explainability regarding why specific transactions are flagged. This lack of interpretability presents a practical challenge for real-world deployment, particularly under strict financial regulations like GDPR that mandate a 'right to explanation' for automated decisions.

B. Data Preprocessing

Before model training, the dataset was found with no missing data and the PCA-transformed features from V1 to V28 were left unchanged. The Time and Amount columns were scaled using Robust Scaler because these two columns contain raw transaction values and are more sensitive to extreme values [3]. The dataset was split into 80% training data and 20% testing data before any resampling was applied. In addition, SMOTE-Tomek was applied only to the training set, while the test set was kept in its original distribution. By doing this, the model was trained with a more balanced dataset but was still tested under realistic fraud detection conditions.

C. Hybrid Resampling using SMOTE-Tomek

Despite partitioning the data into a training set and a testing set, a clear bias occurs because there are more instances for the majority class than the minority class. In an effort to resolve this class imbalance problem, a combined SMOTE-Tomek resampling algorithm is used. Firstly, through SMOTE algorithm, new data points of the minority class can be generated by interpolating between two nearby fraudulent points instead of duplicating the minority data points [1]. This results in an equal amount of both minority and majority classes but the generated minority data points are likely to fall near the majority data points. Secondly, to remove overlapping data points of two different classes, Tomek Links algorithm is used [2]. Although Tomek Links is theoretically designed to clarify class boundaries, PCA-transformed features often already exhibit natural spatial separation. Consequently, the practical necessity and redundancy of this spatial cleaning step on the ULB dataset will be critically evaluated in the subsequent experiments.

D. Cost-Sensitive Random Forest Classification

Random Forest classification algorithm is used in order to classify the dataset. The main characteristic of Random Forest is that it creates an extensive set of decision trees and the prediction is produced by voting. A main drawback associated with the traditional Random Forest is its uniform cost assignment, where equal penalties are assigned to both

types of errors (false positive or false negative). Within the context of credit card fraud detection, not identifying the case of fraud (false negative) results in heavy losses on the financial institution, while false positives may lead only to minor troubles with customers. Therefore, in this case, the introduction of an additional cost matrix is required. After introducing cost matrices, class weights are adjusted in a way that a higher penalty is imposed on a misclassified case of the minority class and false negatives will be punished more than before. Therefore, a new optimization problem arises in which the goal is no longer just maximizing accuracy but rather minimizing financial losses [1].

E. Evaluation Metrics

Standard accuracy is inadequate for evaluating highly imbalanced fraud datasets because it inherently overlooks the rare minority class. Instead, this study utilizes Recall to monitor the detection rate of fraudulent transactions and the F1-Score to balance this detection against operational false alarms. Given the extreme class imbalance, the Precision-Recall Area Under the Curve (PR-AUC) serves as the primary statistical metric, providing a much more robust evaluation of classification performance than traditional ROC-AUC metrics.

In order to match the statistical performance of an algorithm to operational conditions, a cost analysis of finance is embedded into the evaluation method [1]. The use of specific weights for false negatives and costs due to false positives makes it possible to translate the performance of the statistical model into financial losses. In this case, the goal of maximization is not theoretical measures, but practical considerations of achieving maximum profit [2].

An estimated monetary loss metric quantifies the risk based on how much loss each misclassification costs. This ignores the high risk of undetected fraud by imposing a significant cost of Rp 5,000,000 on each False Negative detected from the test. On the other hand, a smaller administrative charge of Rp 50,000 is given to each False Positive for false alarms. The baseline penalty weight of 100 directly reflects the 100:1 mathematical ratio between the operational cost of a false negative (Rp 5,000,000) and a false positive (Rp 50,000). Higher heuristic weights of 200 to 500 were subsequently explored to empirically compensate for the Random Forest's probability calibration bias under extreme class imbalance. Finally, sum up these weighted errors to approximate the total expected monetary loss, giving us a tangible metric of model profitability.

IV. RESULTS AND DISCUSSIONS

A. Early Experimental Results

With regards to the results in the current report, two initial experiments were run to evaluate both the baseline model and the proposed model. The first initial experiment tested the baseline approach that utilized SMOTE with a Random Forest classifier. The baseline model scored 0.8663 on the PR-AUC scale, but was unable to detect 18 cases of real fraud, which resulted in 18 false negatives. Besides, there were 14 instances when the model made a mistake in recognizing legitimate transactions and considered them to be frauds (false positives).

The second initial test was done to examine the performance of the first version of the proposed methodology, namely SMOTE-Tomek with a Cost-Sensitive Random Forest. The results obtained demonstrated slightly lower performance of the model, as the PR-AUC score was 0.8485. The model also had a recall of 0.80 and F1-score of 0.83. During the test, the model was unable to detect 20 cases of fraud, producing 13 false positives during the process.

Because the dataset's PCA transformation naturally separated the classes, Tomek Links found no overlapping rows to remove, rendering the SMOTE-Tomek and SMOTE datasets identical in this experiment. Consequently, our performance comparisons effectively serve as an ablation study highlighting the impact of cost-sensitive penalty tuning rather than spatial cleaning.

B. Performance Comparison and Current Limits

The final scores of both models were compared. Currently, the baseline model exhibits marginally superior performance. The proposed model missed two additional fraudulent transactions compared to the baseline, and the PR-AUC correspondingly decreased from 0.8663 to 0.8485. This slight degradation is anticipated in the initial testing phase, as illustrated in the PR-AUC curve comparison in Fig. 2.

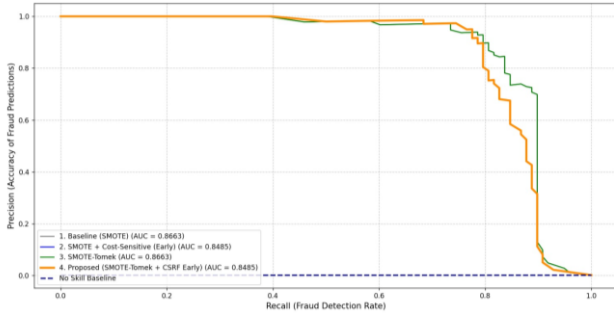


Fig. 2. PR-AUC Curve: A comparison of detection sensitivity and precision-recall trade-offs across all model variants.

In the given chart, there are four curves for different scenarios used for testing. The two curves of the regular SMOTE and the SMOTE-Tomek are very close and have the same value for the PR-AUC, 0.8663. Likewise, the two other curves belonging to the cost-sensitive models are almost identical but have the slightly lower PR-AUC value of 0.8485. One can observe from the graph that all models exhibit a relatively stable precision level up until around a recall of 0.80; thereafter, the precision of the cost-sensitive models drops more rapidly compared to that of the non-cost-sensitive ones. It should be noted that the optimal penalty matrix was not used in the experiment for the Cost-Sensitive Random Forest, and thus the algorithm was not bound by any restrictions concerning minimization of false negatives.

C. Money Loss Analysis and Future Steps

The estimated financial loss was subsequently calculated by assigning specific monetary costs to both false negatives and false positives. For the baseline model, the estimated financial liability amounted to Rp 90,700,000. Conversely, the proposed model incurred a higher estimated loss of Rp 100,650,000, primarily driven by the 20 undetected

fraudulent transactions, as detailed in the confusion matrices in Fig. 3.

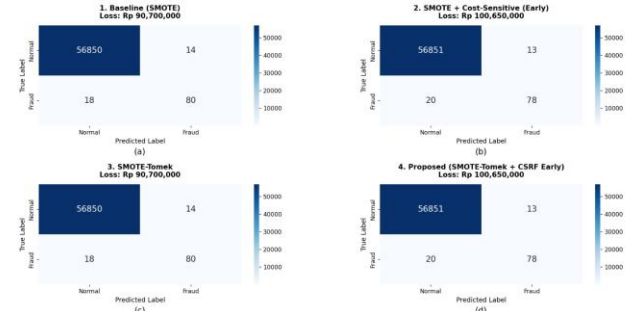


Fig. 3. Confusion Matrix and Financial Loss: (a) SMOTE and Random Forest, (b) SMOTE and Cost-Sensitive Random Forest, (c) SMOTE-Tomek and Random Forest, (d) SMOTE-Tomek and Cost-Sensitive Random Forest

The impact of these results is fully detailed in the Confusion Matrix comparisons, where model predictions are directly translated into these financial figures. The matrices top-left and bottom-left show that the Baseline (SMOTE) and SMOTE-Tomek models produced 18 False Negatives (missed frauds) and 14 False Positives (wrongly blocked payments). The right-side matrices, on the other hand, show that the SMOTE + Cost-Sensitive and the Proposed SMOTE-Tomek + Cost-Sensitive models recorded 20 False Negatives and 13 False Positives, respectively.

The false alarms were reduced to 13 cases but the cost of missed frauds is much higher. While the number of false alarms was successfully reduced by the proposed model by one case compared to the baseline, two additional actual fraud cases were incorrectly classified as normal. Because the financial penalty for a False Negative is significantly larger than the administrative cost of a False Positive, this slight increase in missed frauds directly caused the total estimated loss to surge. While the fixed Rp 50,000 penalty quantifies direct administrative overhead, it inherently simplifies the critical trade-off between operational efficiency and customer experience. Future iterations must account for 'customer friction,' as prioritizing fraud prevention through increased false alarms risks alienating high-value customers and incurring unquantified reputational costs.

D. Fine-Tuning Results for Minimized Monetary Loss

After the first experiment, an additional fine-tuning step was performed to select the best cost-sensitive penalty weights for the minority class. To evaluate the practical efficiency of the model, an estimated financial impact is computed by assigning a much larger penalty weight to each False Negative and a relatively small penalty to each False Positive. The sum of these weighted classification errors yields a total penalty score that directly reflects the actual financial risks associated with prediction errors. A detailed breakdown of the model performance and financial impacts across different penalty weights is presented in Table 1.

TABLE I. COST-SENSITIVE PENALTY OPTIMIZATION RESULTS

Penalty Weight (0:1, 1:X)	PR-AUC Score	Missed Frauds (FN)	False Alarms (FP)	Total Financial Loss
100	0.8485	20 cases	13 cases	Rp 100,650,000

200	0.8517	19 cases	15 cases	Rp 95,750,000
203	0.8505	20 cases	14 cases	Rp 100,700,000
204	0.8526	19 cases	14 cases	Rp 95,700,000
205	0.8534	20 cases	15 cases	Rp 100,750,000
206	0.8528	20 cases	14 cases	Rp 100,700,000
207	0.8534	20 cases	14 cases	Rp 100,700,000
208	0.8541	19 cases	14 cases	Rp 95,700,000
209	0.8549	19 cases	14 cases	Rp 95,700,000
500	0.8475	20 cases	14 cases	Rp 100,700,000

As seen from Table 1, any penalty weights that were less than 200 were unable to improve on the baseline loss of Rp 100.65 million. The ideal weighting would occur for the interval of weights between 204–209, which provided the smallest financial loss of Rp 95.7 million along with missing the fewest number of fraud cases at 19. Any weight that exceeded 209, for instance 500, produced a decline in PR-AUC to 0.8475.

E. Final Model Comparison

After the penalty weight was set to 204, the SMOTE-Tomek with Cost-Sensitive Random Forest and SMOTE and Cost-Sensitive Random Forest model showed a better performance. In this setting, the proposed model PR-AUC score reached 0.8526, while the number of fraud cases that were not detected was reduced to 19 cases. In the previous experiment with default or lower weights, 20 fraud cases were still missed and the estimated loss was Rp 100,650,000. After the weight was adjusted, the total loss was reduced to Rp 95,700,000. From the confusion matrix, it can also be seen that the model became more balanced in finding fraud cases, but it still did not create too many false alarms. The False Positive value was 14 cases, which is still acceptable for this model. Because of this, the weight of 204 can be considered as a better setting, since it reduced the financial loss without making the model too sensitive to fraud prediction. A complete comparison of the financial loss across all models is summarized in Table II, whereas the corresponding PR-AUC performance curves for the four experimental configurations are illustrated in Fig. 4.

TABLE II. MODEL COMPARISON FINANCIAL LOSS

Model	TN	FP	FN	TP	Financial Loss
SMOTE	56.850	14	18	80	Rp 90,700,000
SMOTE-Tomek	56.850	14	18	80	Rp 90,700,000

SMOTE + CSRF	56.850	14	19	79	Rp 95,700,000
SMOTE-Tomek + CSRF	56.850	14	19	79	Rp 95,700,000

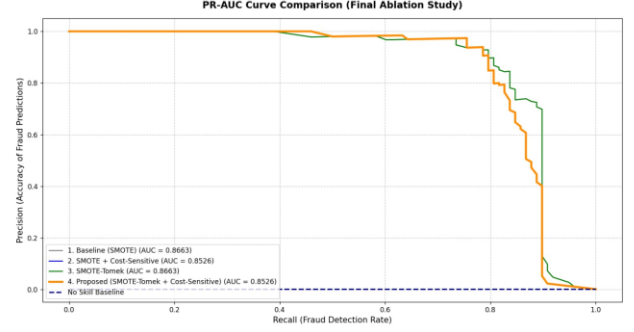


Fig. 4. PR-AUC Curve: A final comparison of detection sensitivity and precision-recall trade-offs across all model variants.

The PR-AUC performance of the four experimental configurations shows that the baseline SMOTE-RF and SMOTE-Tomek-RF models resulted in the highest PR-AUC value of 0.8663, which suggests a slightly better overall precision–recall trade-off than the cost-sensitive variants. On the other hand, the introduced SMOTE-Tomek and Cost-Sensitive Random Forest, i.e. both cost-sensitive models, achieved a PR-AUC of 0.8526 after the optimization of penalty weights. Although this is a small gain in PR-AUC, the cost-sensitive method was designed to minimize the financial impact of missed fraud cases only, not to optimize a statistical metric. Therefore, the PR-AUC curves demonstrate that the proposed model preserves a competitive detection capability while achieving a more favorable balance between fraud detection performance and financial risk reduction.

To answer the first research question, we must address why the SMOTE and SMOTE-Tomek baseline models produced identical results. Both models achieved the exact same PR-AUC score and missed the same number of frauds. A check on the training dataset size explains this outcome. The basic SMOTE method generated a balanced set of 454,902 rows. Following this, the Tomek Links algorithm removed zero rows from the dataset. This occurred because the dataset uses PCA (Principal Component Analysis) features. The PCA transform naturally discriminates between different classes of data in a multidimensional space. Therefore, the synthetic fraud points produced by SMOTE were outside the normal transaction boundaries. Since Tomek Links specifically targets overlapping data pairs, it found nothing to clean. The Random Forest models trained on the exact same data rows, forcing them to make identical predictions.

V. CONCLUSION

The goal of this research is to reduce financial loss in credit card fraud detection with the combination of SMOTE-

Tomek cleaning method and Cost-Sensitive Random Forest. The results showed that changing the penalty weight to 204 can minimize missed fraud cases to 19 cases and minimize the estimated loss to Rp 95,700,000. However, the evaluation also showed a major constraint, the original SMOTE and SMOTE-Tomek Random Forest models still obtained the lowest overall loss (Rp 90,700,000) on this particular data. This implies that the hybrid architecture proposed is effective only on the exact topology of the dataset. Due to the PCA transformation of the features, the data classes are orthogonalized in a natural way and the spatial boundary cleaning of Tomek Links is not needed, reducing the overall impact of the cost-sensitive approach.

In future work, researchers should attempt to validate this cost-sensitive framework on larger, raw (non-PCA) transaction datasets, where class overlapping is naturally more extreme. Moreover, the use of dynamic cost matrices that adapt to different real-time transaction volumes can further optimize penalty weights. This would be a more robust and flexible risk-control mechanism for financial institutions dealing with dynamic fraud patterns.

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AUTHOR CONTRIBUTION STATEMENT

Jason Benoit Adianto: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing - Review & Editing. **Radianda Setiawan:** Methodology, Software, Validation, Investigation, Writing - Review & Editing. **Ivan Sebastian Edbert:** Supervision. **Samson Ndruru:** Supervision, Conceptualization, Validation, Writing - Review & Editing.

DATA AVAILABILITY

The data for this study is publicly available at the following link:

<https://www.kaggle.com/datasets/mlgulf/creditcardfraud>

AI USAGE DECLARATION

We declare that we used artificial intelligence (AI) tools, specifically Gemini, which was developed by Google, to prepare this manuscript. These tools were used for tasks such as grammar correction, language refinement, and paraphrasing suggestions. All AI-generated content was reviewed, edited, and validated by the authors to ensure accuracy, clarity, and adherence to academic standards.

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